

ZIRUI YAN

AI Research Engineer II, Duolingo

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Research Interests

Causal bandits, machine learning, and large language models.

Industry Experience

Duolingo

AI Research Engineer II, Monetization Eigen Team

New York, NY

Jun 2026 - Present

- Working on extending bandit to mid-session no energy case to boost the subscription

Duolingo

AI Research Engineer Intern, Monetization Eigen Team (Mentor: Badr Albanna)

Pittsburgh, PA

May 2025 - Aug 2025

- Accelerated training and deployment pipeline by >40% on **BigQuery and Vertex AI** in Google Cloud
- Pre-processed billions of ad history records using **Polars** for v2 experiment in <1h
- Added **temporal features** that capture ad interactions and event-time dependence
- Launched **new ads bandit experiment**, boosting session-end conversion rate by 10% (\$35k/day in revenue)

Yahoo! Research

Research Intern, Search and Recommendation

Remote

(Mentor: Xinyue Wang, Manager: Rao Shen)

May 2024 - Aug 2024

- Worked on Yahoo! search query understanding for downstream tasks
- Generated pseudo-labels with >90% accuracy via **zero-shot prompting**
- Achieved >70% knowledge-transfer accuracy to **compact BERT models** through data augmentation

IBM Research

Research Extern, Trustworthy AI (Mentor: Tian Gao)

Yorktown Heights, NY

May 2023 - Aug 2023

- Developed efficient **bandit-based prompt selection** algorithms for large language models (LLM)
- Addressed non-stationary, stochastic reward settings, and outperformed RL baselines by 10% in LLMs

Education

Rensselaer Polytechnic Institute

Ph.D., Electrical, Computer, and Systems Engineering

Troy, NY

2021 - 2026

Dissertation: *Causal Bandits with Soft Intervention: A Unified Framework*. Advisor: Prof. Ali Tajer.

University of Science and Technology of China

B.S. in Statistics and B.E. in Computer Science and Technology

Hefei, China

2016 - 2020

Peer-Reviewed Publications (* equal contribution)

- Causal Bayesian Optimization via Causal Bandits: Open Questions
A. Mukherjee, **Z. Yan**, A. Tajer
Proc. UAI Workshop on Causality for Decision Making (**UAI CDM**) 2026.
- Multi-component Causal Tracing in Large Language Models
Z. Yan, D. Wei, D. A. Katz, P. Sattigeri, A. Tajer
Proc. Association for Computational Linguistics (**ACL**) 2026. Oral presentation.

Peer-Reviewed Publications (continued)

- Reward-oriented Causal Representation Learning
Z. Yan*, E. Acartürk*, A. Tajer
Proc. Conference on Neural Information Processing Systems (**NeurIPS**) 2025
- Linear Causal Bandits: Unknown Graph and Soft Interventions
Z. Yan, A. Tajer
Proc. Conference on Neural Information Processing Systems (**NeurIPS**) 2024
- Improved Bound for Robust Causal Bandits with Linear Models
Z. Yan, A. Mukherjee, B. Varıcı, A. Tajer
Proc. International Symposium on Information Theory (**ISIT**) 2024
- Causal Bandits with General Causal Models and Interventions
Z. Yan, D. Wei, D. A. Katz, P. Sattigeri, A. Tajer
Proc. International Conference on Artificial Intelligence and Statistics (**AISTATS**) 2024
- Robust Causal Bandits for Linear Models
Z. Yan, A. Mukherjee, B. Varıcı, A. Tajer
IEEE Journal on Selected Areas in Information Theory (**JSAIT**) 2024
- Federated multi-armed bandit via uncoordinated exploration
Z. Yan, Q. Xiao, T. Chen, A. Tajer
Proc. International Conference on Acoustics, Speech, and Signal Processing (**ICASSP**) 2022
- Optimizing Parameter Mixing Under Constrained Communications in Parallel Federated Learning
X. Liu*, **Z. Yan***, Y. Zhou, D. Wu, X. Chen, J. H. Wang
IEEE/ACM Transactions on Networking (**TON**), 2023
- Image denoising via K-SVD with primal-dual active set algorithm
Q. Xiao, C. Wen, **Z. Yan**
Proc. Winter Conference on Applications of Computer Vision (**WACV**), 2020

Professional Service

- Conference reviewer: ICML (2025), ICLR (2025), NeurIPS (2024–2026), AISTATS (2024), and ISIT; NeurIPS 2024 Top Reviewer.
- Journal reviewer: IEEE Transactions on Signal Processing, Transactions on Machine Learning Research (2025), and IEEE Journal on Selected Areas in Information Theory.
- Workshop reviewer: ICML 2023 Workshop on Structured Probabilistic Inference and Generative Modeling (SPIGM).

Technical Skills

Programming: Python, SQL, R, LaTeX, Markdown

Machine Learning and Data: PyTorch, vLLM, Hugging Face, pandas, Polars, DBT, Vertex AI, BigQuery

Systems: Docker, Kubernetes